Question 3 0 out of 0.5 points

If a suitable comparison sort algorithm is used in Radix sort, then the time complexity will not be linear.

Selected Answer: ✗ False

Answers: ✓ True

False

Question 4 0 out of 0.5 points

The difference between merge sort and heap sort is that:

Selected Answer: ✗ Merge sort is faster than the heap sort.

Answers: ✓ Heap sort is in-place while the merge sort is not.

Merge sort is faster than the heap sort.

Heap sort is stable while merge sort is not.

All the above answers are correct.

0 out of 0.5 points

Question 7

0.5 out of 0.5 points

The correct sorting steps when applying insertion sort on the array A[3,6,2,1] are:

Selected Answer: [3,6,2,1] -> [3,6,2,1] -> [2,3,6,1] then [1,2,3,6]

Answers:

[3,6,2,1] -> [1,3,6,2] -> [1,2,3,6] then [1,2,3,6]

[3,6,2,1] -> [6,2,3,1] -> [6,1,2,3] then [1,2,3,6]

[3,6,2,1] -> [1,6,2,3] -> [1,2,6,3] then [1,2,3,6]

[3,6,2,1] -> [3,6,2,1] -> [2,3,6,1] then [1,2,3,6]

Question 8

0.5 out of 0.5 points

It is suitable to apply ____ sort if the inputs are distributed uniformly over [0, 1).

Selected Answer: Bucket

Answers:

Insertion

Radix

Bucket

Counting

Question 5

0 out of 0.5 points

Which of the following uses the largest amount of space for sorting?

Selected Answer: ✗ Insertion sort

Answers:
- Heapsort
- Insertion sort
- ✓ Merge sort
- All the above algorithms are the same

Question 6

0 out of 0.5 points

The asymptotically tight upper bound of building the max heap is:

Selected Answer: ✗ All the above answers are incorrect

Answers:
- ✓ O(n)
- O(n logn)
- O(log n)
- All the above answers are incorrect

Question 9

0 out of 0.5 points

To sort a list of numbers in descending order (from the largest to the smallest), we use Max heap.

Selected Answer: ✗ True

Answers: True

✓ False

Question 10

0.5 out of 0.5 points

The best algorithm used to sort a nearly sorted list is:

Selected Answer: ✓ Insertion sort

Answers: Merge sort

Heap sort

✓ Insertion sort

All the above answers are correct

Sunday, Sha'bān 8, 1442 3:05:55 PM AST

← OK

Question 1

0 out of 0.5 points

To implement radix sort, _____ sort can be used.

Selected Answer: Both (a) and (b) are incorrect.

Answers:
- Heap
- Merge
- Both (a) and (b) are correct.
- Both (a) and (b) are incorrect.

Question 2

0.5 out of 0.5 points

Using the suitable searching algorithm, searching for a specific item in an ordered list of *n* items, you may need to apply *n* comparisons to find the target.

Selected Answer: False

Answers:
- True
- False

Question 3

0 out of 0.5 points